

SAUMALKOL VOLODARSKOYE, Kazakhstan

WMO: 288770

Lat: 53.317N

Lon: 68.100E

Elev: 1047

StdP: 14.15

Time Zone: 6.00 (E06)

Period: 83-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.3	-17.9	-28.7	1.1	-24.1	-22.7	1.6	-17.8	37.0	8.3	32.1	19.5	4.3	230	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	86.0	65.0	82.8	64.8	79.4	63.3	69.0	79.9	67.0	77.7	65.4	75.9	6.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	96.2	74.8	63.0	89.6	72.5	61.2	84.0	70.9	33.8	79.7	32.1	77.6	30.8	76.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 27.2	22.4	19.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-27.8	71.5	6.7	2.2	-32.6	73.1	-36.5	74.4	-40.2	75.6	-45.0	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	6.5	4.4	16.8	37.5	53.0	64.5	66.3	61.3	51.1	37.6	16.3	9.7		
	DBStd	24.72	11.33	10.95	9.74	11.17	9.90	9.09	6.59	6.21	8.50	9.06	12.11	11.23		
	HDD50	6895	1348	1277	1028	402	86	11	0	1	87	394	1010	1251		
	HDD65	10989	1813	1697	1493	824	386	116	64	148	422	850	1460	1716		
	CDD50	1638	0	0	0	27	178	447	506	352	119	9	0	0		
	CDD65	259	0	0	0	0	13	102	106	34	4	0	0	0		
	CDH74	2259	0	0	0	1	147	1017	823	220	51	0	0	0		
	CDH80	601	0	0	0	0	18	322	226	33	2	0	0	0		
	WSAvg	7.3	9.6	8.9	6.7	7.8	7.1	5.8	5.4	5.4	6.7	7.1	7.6	9.3		
	PrecAvg	16.8	0.9	0.7	0.9	1.1	1.3	1.7	3.0	2.0	1.3	1.4	1.3	1.1		
(15) Precipitation	PrecMax	21.7	1.9	1.5	2.0	3.6	2.1	3.3	6.2	4.9	2.5	2.3	2.5	2.3		
	PrecMin	10.5	0.3	0.1	0.2	0.0	0.1	0.5	1.1	0.3	0.4	0.2	0.2	0.5		
	PrecStd	3.3	0.4	0.4	0.5	0.9	0.6	0.8	1.6	1.5	0.6	0.6	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	29.6	38.7	71.6	82.2	91.6	90.4	84.2	79.5	66.7	40.5	34.8		
		MCWB	30.6	28.4	33.9	53.2	59.8	64.3	69.9	66.3	59.3	50.7	37.3	33.0		
	2%	DB	29.7	27.3	35.0	66.1	78.0	87.2	85.3	80.0	74.7	58.9	35.1	30.4		
		MCWB	28.6	26.2	32.3	51.2	58.1	64.5	67.6	64.8	58.2	47.0	32.3	29.6		
	5%	DB	25.8	24.3	32.8	60.5	74.1	84.0	82.0	76.4	69.3	54.5	33.1	27.6		
		MCWB	24.7	22.8	30.7	48.7	56.1	63.8	66.5	62.9	55.2	44.6	31.4	26.5		
	10%	DB	22.9	20.7	30.5	54.6	69.9	80.2	78.6	73.0	64.9	50.7	31.0	25.1		
		MCWB	21.6	19.1	28.5	46.3	54.9	62.9	65.2	62.0	53.5	43.2	29.4	24.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	28.6	35.6	55.1	62.1	69.9	72.7	69.6	64.8	52.6	37.6	33.6		
		MCDB	31.8	29.2	37.7	66.5	76.8	84.1	85.7	77.4	75.4	62.8	40.7	34.2		
	2%	WB	28.5	26.2	32.8	52.1	60.2	67.3	70.2	66.5	59.3	49.1	33.2	29.8		
		MCDB	29.9	27.1	34.4	63.7	74.1	81.0	81.4	75.8	70.6	57.2	34.2	30.4		
	5%	WB	24.8	22.9	30.7	49.4	57.8	65.4	68.2	64.7	56.4	45.8	31.3	26.7		
		MCDB	25.6	24.3	32.6	60.4	70.4	79.8	78.2	73.3	66.5	52.9	32.6	27.6		
	10%	WB	21.8	19.3	28.6	46.2	55.7	63.6	66.4	63.0	54.0	43.3	29.3	24.0		
		MCDB	22.8	20.8	30.4	54.1	68.3	77.8	76.4	71.4	63.7	50.2	30.8	25.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	13.8	15.0	15.5	19.4	19.5	16.5	16.1	17.2	14.2	12.8	13.0		
		MCDBR	14.2	11.5	14.2	22.7	25.5	23.7	21.5	22.0	25.1	20.9	12.5	14.9		
		MCWBR	14.0	10.9	12.2	12.3	12.2	10.5	9.0	10.3	12.9	11.5	11.6	14.3		
		MCDBR	13.2	11.3	13.2	22.3	22.7	21.2	19.1	19.2	22.1	19.2	13.0	14.1		
	5% WB	MCWBR	13.1	11.2	11.9	13.5	12.1	11.1	9.5	10.3	13.4	12.0	12.4	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.311	0.365	0.401	0.399	0.400	0.420	0.416	0.365	0.345	0.315	0.281		
	taud		2.076	2.042	1.984	2.143	2.235	2.309	2.243	2.236	2.335	2.278	2.140	2.057		
	Ebn at Noon		210	242	252	261	269	270	261	254	255	231	193	182		
	Edh at Noon		23	33	44	43	41	39	41	39	31	26	22	19		
(44) All-Sky Solar Radiation	RadAvg		370	724	1192	1471	1801	1925	1776	1508	1087	574	282	246		
	RadStd		27	32	126	154	148	155	138	130	79	84	40	31		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page